Do Language Models Converge to Themselves? Recursive Self-Refinement as Textual Relaxation

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Abstract

Large language models are increasingly used in recursive refinement workflows, where an initial draft is repeatedly revised by the same model. Despite the growing use of such workflows, their long-term dynamical behavior remains largely unexplored. Does repeated refinement continue to improve outputs indefinitely, or does it converge toward a stable textual form? We study recursive self-refinement as a dynamical process, in which repeated LLM revision drives text toward a model-preferred soft fixed-point region. Using GPT-5.5, we generate refinement trajectories on 50 ICML 2025 abstracts over 10 iterations under both default-temperature and deterministic decoding, and additionally evaluate 15 ICML 2020 abstracts as a cross-year comparison. We characterize these trajectories using normalized edit distance, exact and approximate fixed-point statistics, word-count stability, exponential relaxation analysis, and external LLM-as-a-judge evaluation. Across settings, refinement trajectories exhibit rapid saturation. Most edits occur within the first few iterations, after which trajectories enter a soft fixed-point region characterized by only small residual surface-level changes. Deterministic decoding reaches exact fixed points more rapidly and exhibits substantially smaller residual fluctuations than default-temperature decoding, while both settings achieve universal approximate convergence. The average edit magnitude $\Delta_t = d(V_t, V_{t+1})$ follows a consistent exponential relaxation pattern, suggesting a relaxation process toward a model-preferred textual equilibrium rather than open-ended optimization. External evaluation on sampled trajectories indicates that converged abstracts improve clarity, conciseness, and scientific style while preserving technical meaning. These results support a dynamical-systems view of LLM self-refinement and motivate practical stopping criteria based on edit-magnitude saturation.

1 Introduction

Large language models (LLMs) are increasingly used not only to generate text, but also to revise their own outputs. In many practical workflows, a user or system provides an initial draft, asks the model to improve it, and then feeds the revised version back into the same model for further refinement. Such recursive self-refinement is common in scientific writing, coding, summarization, and agentic systems. Yet despite its growing use, its dynamical behavior remains poorly understood. Does repeated refinement continue to improve a text indefinitely, or does it relax toward a stable textual form?

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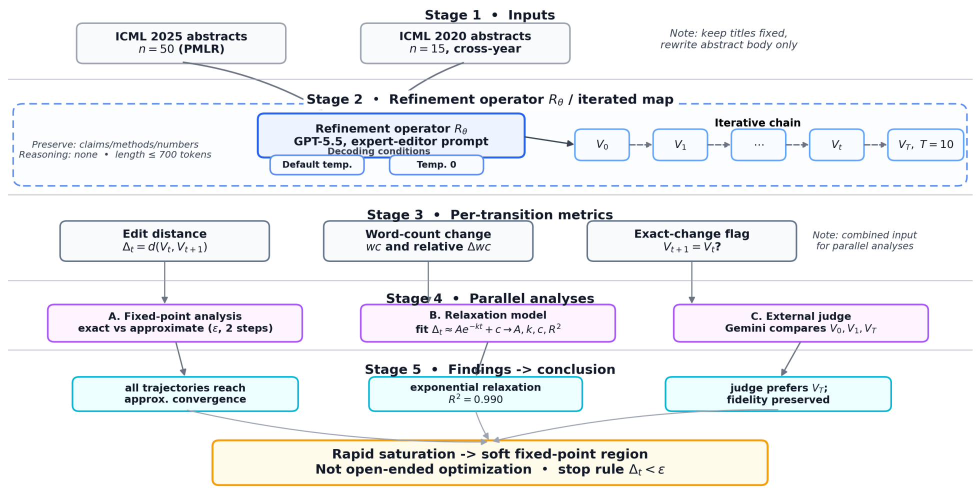

Figure 1: Overview of recursive self-refinement as a discrete dynamical system over text. Starting from an original abstract V_0, the same refinement operator R_θ is repeatedly applied to generate a trajectory $V_0, V_1, \ldots, V_T$. We measure transition magnitudes $\Delta_t = d(V_t, V_{t+1})$, analyze exact and approximate fixed points, and fit an exponential relaxation model $\Delta_t \approx Ae^{-kt} + c$. The residual term c captures a soft fixed-point floor, where further refinements produce only small surface-level fluctuations. An external Gemini judge compares V_0, V_1, and V_T to assess whether convergence corresponds to quality improvement rather than mere textual stagnation.

Understanding recursive self-refinement is important for both practical and scientific reasons. Practically, users need to know how many refinement rounds are useful before additional iterations become redundant or potentially harmful. Scientifically, recursive self-refinement defines a discrete dynamical system over text: each model call maps one document version to another. This perspective raises natural questions about convergence, stability, attractors, residual fluctuations, and stopping criteria. However, most existing evaluations of LLM editing focus on single-step quality improvements rather than the dynamics induced by repeated self-application.

In this work, we study recursive self-refinement as *textual relaxation*: a process in which repeated LLM revision drives an initial draft toward a model-preferred soft fixed-point region. Unlike open-ended optimization, textual relaxation predicts that edit magnitudes should rapidly decrease and eventually approach a small residual floor. This framing connects practical refinement workflows to dynamical-systems concepts such as fixed points, convergence rates, saturation, and stability under repeated self-application.

We empirically examine this view in the domain of scientific abstract revision, where improvements in clarity and style must preserve technical meaning. Starting from ICML paper abstracts, we repeatedly ask GPT-5.5 to revise each abstract while preserving all claims, methods, results, and numerical values. This produces a trajectory of versions $V_0, V_1, \ldots, V_T$, where V_0 is the original abstract and V_{t+1} is the model's refinement of V_t. We analyze these trajectories using normalized edit distance, word-count stability, exact fixed-point statistics, approximate convergence criteria, exponential relaxation fitting, and external LLM-as-a-judge evaluation.

A central challenge is that exact textual fixed points are too strict for natural language refinement. Even after a text is essentially stable, the model may continue to make small surface-level changes, such as punctuation, hyphenation, or minor phrasing adjustments. We therefore distinguish exact fixed points, where $V_{t+1} = V_t$, from approximate fixed points, where consecutive versions differ only by a small edit distance and exhibit stable length. This distinction allows us to characterize recursive refinement as convergence to a soft fixed-point region rather than to a single canonical string.

Empirically, we find that recursive self-refinement converges rapidly. On 50 ICML 2025 abstracts refined for 10 iterations, GPT-5.5 reaches approximate convergence for all abstracts under both default-temperature and deterministic decoding. Under temperature 0, all abstracts also reach exact textual fixed points, with a mean exact convergence time of 2.6 iterations. The average transition magnitude $\Delta_t = d(V_t, V_{t+1})$ decays sharply over time and is well described by an exponential relaxation curve, suggesting a relaxation process toward a model-preferred textual equilibrium rather than indefinite iterative optimization. Default-temperature decoding exhibits the same qualitative convergence behavior, but with larger residual fluctuations and later exact fixed points.

We further validate that convergence is not merely textual stagnation. An external Gemini judge evaluates sampled trajectories and consistently prefers the final refined abstracts over the original and first-refined versions, while assigning perfect technical-fidelity scores across versions. This suggests that recursive self-refinement improves clarity, conciseness, and scientific style without sacrificing technical meaning. A smaller cross-year experiment on ICML 2020 abstracts shows similar convergence behavior, indicating that the observed dynamics are not unique to contemporary LLM-era scientific writing.

Our contributions are threefold. First, we frame recursive LLM self-refinement as textual relaxation: a discrete dynamical process over text that approaches a model-preferred soft fixed-point region. Second, we provide empirical evidence that refinement trajectories rapidly saturate, with edit magnitudes following a regular exponential relaxation pattern. Third, we show that decoding temperature controls residual fluctuations around this region, and that edit-magnitude saturation motivates practical stopping criteria for recursive revision workflows. Together, these findings support a dynamical-systems view of LLM revision and suggest that iterative AI systems can be studied through the lens of convergence, stability, and textual equilibrium.

2 Related Work

Recursive self-refinement and reflection. Large language models are increasingly used in iterative refinement workflows, where an initial output is repeatedly revised through self-critique, reflection, or feedback. Self-Refine [Madaan et al., 2023] introduced a framework in which a model generates feedback on its own outputs and iteratively revises them. Reflexion [Shinn et al., 2023] extended this idea to language agents that accumulate verbal feedback across attempts. Constitutional AI [Bai et al., 2022] similarly employs self-critique and revision guided by explicit principles. Collectively, these studies demonstrate that iterative refinement can improve output quality across diverse tasks. However, their primary focus is performance improvement after refinement, rather than the long-term dynamics induced by repeatedly applying the same model to its own outputs.

LLM revision and evaluation. Text revision systems are commonly evaluated using human judgments, semantic similarity measures, and automated metrics. More recently, strong language models have been shown to function as effective evaluators of generated text [Zheng et al., 2023, Liu et al., 2023]. Such LLM-as-a-judge approaches are increasingly used to assess qualities such as clarity, helpfulness, style, and factual consistency. Scientific abstract refinement is a demanding setting because stylistic improvements must preserve technical meaning. We therefore combine trajectory-level convergence metrics with external LLM-based evaluation of clarity, conciseness, scientific style, and technical fidelity. This allows us to distinguish genuine refinement from mere textual stagnation.

Iterative reasoning and inference-time computation. Recent work has shown that repeated inference-time computation can improve language-model performance. Examples include Chain-of-Thought prompting [Wei et al., 2022], Self-Consistency [Wang et al., 2023], Tree-of-Thoughts [Yao et al., 2023], and related inference-time scaling approaches. These methods explore, sample, evaluate, or revise intermediate reasoning paths to improve final-task accuracy. However, their primary objective is better task performance, not characterizing the trajectory produced by repeated self-application. In contrast, our work focuses on refinement trajectories themselves and asks whether repeated revision converges toward stable textual equilibria.

Dynamical systems, fixed points, and relaxation. Many iterative computational processes can be analyzed through the lens of dynamical systems, fixed points, convergence rates, attractors, and stability regions. Classical iterative algorithms are often characterized by how rapidly their residuals decay and whether they approach stable equilibria. Similar questions naturally arise in recursive LLM